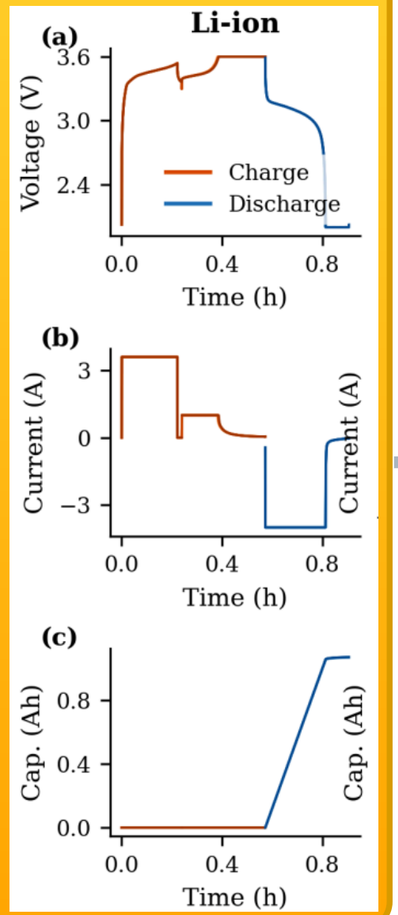
A Multi-Chemistry Battery Degradation Data

Li-ion

Na-ion

Zn-ion

CALB



B Preprocessing

Resampling
300 points/cycle

Normalization
Per-battery scaling

Padding & Masking
Fixed length (L=100)

Sequence Construction
100-cycle sequence

C Proposed Tiny AI Framework

TinyBatteryNet

Compact On-Device AI Model

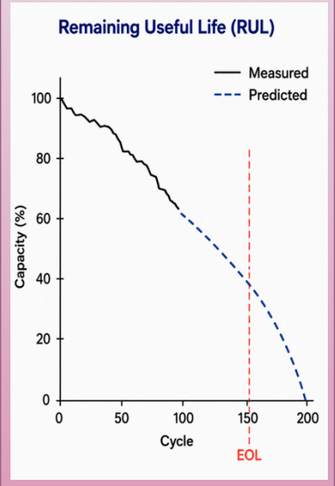
Lightweight & Compact

Real-Time Inference

Low Memory Footprint

Designed for Edge Devices

D RUL Prediction (On-Device)



Prediction Error Analysis

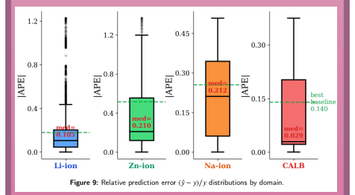


Figure 9: Relative prediction error $(\hat{y} - y)/y$ distributions by domain.

E On-Device Deployment

Battery System

Microcontroller
(STM32F4 Series)

On-Device
RUL Estimation

F Applications Enabled by TinyBatteryNet

Electric Vehicles (EVs)

RUL tracking
Maintenance planning
Range reliability

Energy Storage Systems (ESS)

Health monitoring
Risk management
Lifecycle optimization

Industrial Equipment & Robotics

Predictive maintenance
Downtime reduction
Productivity improvement

IoT & Wireless Sensor Nodes

Autonomous health assessment
Long device lifetime

Second-Life Batteries



RUL assessment
Repurposing selection
Value maximization